

# Boolean functions:

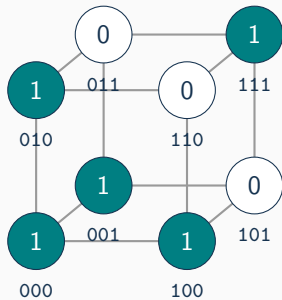
## applications in quantum error correction and beyond

The same structure can recover a message, expose a predictor, and control a gate.

1 Recover a message      2 Understand a rule      3 Control a gate

Bohan Lu   Duke University

## A damaged message can still contain an exact rule



A sender evaluates a simple binary rule at the eight vertices of a cube.

One transmitted value is flipped.

The receiver knows the rule was affine: an XOR of selected input bits, possibly plus a constant.

Can we recover the message without knowing which entry is wrong?

## A Boolean function becomes a transmitted word

A Boolean function maps  $u \in \mathbb{F}_2^3$  to a bit. Our original rule is

$$p(u) = u_1 + u_2 + u_3.$$

| $u$             | 000 | 001 | 010 | 011 | 100 | 101 | 110 | 111 |
|-----------------|-----|-----|-----|-----|-----|-----|-----|-----|
| $p(u)$          | 0   | 1   | 1   | 0   | 1   | 0   | 0   | 1   |
| received $r(u)$ | 1   | 1   | 1   | 0   | 1   | 0   | 0   | 1   |

$$\text{wt}(p) = 4, \quad \text{wt}(r) = 5, \quad d(p, r) = \text{wt}(p + r) = 1.$$

Function addition is in  $\mathbb{F}_2$ :  $1 + 1 = 0$ . Weights and distances are ordinary integer counts.

## ANF turns a truth table into a polynomial

Every Boolean function has a unique *algebraic normal form*: a sum of square-free monomials over  $\mathbb{F}_2$ .

$$p = u_1 + u_2 + u_3, \quad \deg p = 1.$$

The indicator of the flipped vertex is

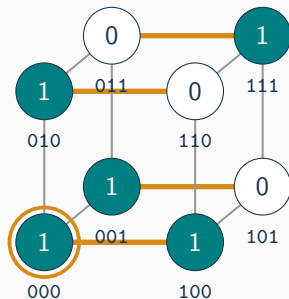
$$\delta_{000}(u) = (1 + u_1)(1 + u_2)(1 + u_3).$$

Therefore the received table has polynomial

$$r = p + \delta_{000} = 1 + u_1u_2 + u_1u_3 + u_2u_3 + u_1u_2u_3.$$

One flipped table entry raised the degree from 1 to 3, but the original codeword is still only one entry away.

## Boolean derivatives turn edges into consistency checks



Let  $e_1 = (1, 0, 0)$ . Define

$$D_{e_1}f(u) = f(u + e_1) + f(u).$$

The derivative is 1 when the endpoints disagree.

$$D_{e_1}p = 1,$$

$$D_{e_1}r = 1 + (1 + u_2)(1 + u_3).$$

An affine function has constant derivatives. One error spoils one of the four disjoint edge checks.

## Local votes recover all four message bits

For an affine rule,

$$f_{a,b}(u) = b + a_1u_1 + a_2u_2 + a_3u_3, \quad D_{e_i}f_{a,b} = a_i.$$

| $e_1$ edge   | 000–100 | 001–101 | 010–110 | 011–111 |
|--------------|---------|---------|---------|---------|
| endpoint sum | 0       | 1       | 1       | 1       |

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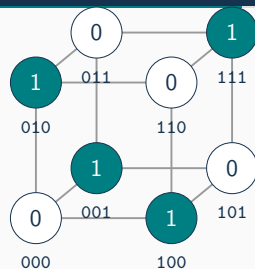
| $e_1$ edge   | 000–100 | 001–101 | 010–110 | 011–111 |
|--------------|---------|---------|---------|---------|
| endpoint sum | 0       | 1       | 1       | 1       |

Majority gives  $a_1 = 1$ . The other directions give  $a_2 = a_3 = 1$ .

Adding the recovered parity to  $r$  leaves seven zeros and one one: majority gives  $b = 0$ .

The decoder recovers the exact message using only the received word and the degree-one promise.

## Affine rules split binary space into equal halves

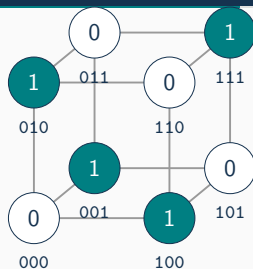


An *affine hyperplane* is a level set of a nonconstant affine rule.

$$H = \{u : p(u) = 1\} = \{100, 010, 001, 111\}.$$

Four ones; four zeros.

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$$H = \{u : p(u) = 1\} = \{100, 010, 001, 111\}.$$

Four ones; four zeros.

For distinct affine rules,

$$d(f, g) = \text{wt}(f + g) \in \{4, 8\}.$$

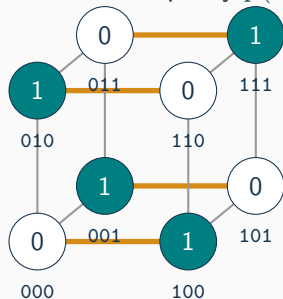
A nonconstant difference has four ones; the constant 1 has eight.

RM(1, 3) is an  $[8, 4, 4]$  code: eight bits, four message coefficients, distance four.

Unique one-error correction.

## Changing one input can change every output

Return to the clean parity  $p(u) = u_1 + u_2 + u_3$ , with uniformly random inputs.



**Observe one input:** the output is still equally likely to be 0 or 1.

**Flip that input:** the output always changes.

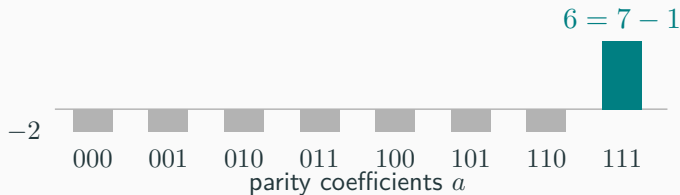
$$D_{e_1}p(u) = 1 \quad \text{at every vertex.}$$

Boolean derivatives identify which input changes reach a digital circuit's output. Sensitivity and predictability ask different questions.

## Fourier analysis counts agreement with simple rules

Represent outputs by real signs. The Walsh/Fourier score of a parity is

$$W_f(a) = \sum_{u \in \mathbb{F}_2^3} (-1)^{f(u) + a \cdot u} = \text{agreements} - \text{disagreements}.$$



For our received word  $r$ , the peak identifies  $p$ : seven agreements survive one error.

Unnormalized Walsh transform; O'Donnell, Analysis of Boolean Functions.

## A useful decoder can reveal an unwanted predictor

The same table  $r$  has a cubic ANF, yet agrees with parity on seven of eight inputs:

$$\deg r = 3, \quad \Pr_{u \text{ uniform}} [r(u) = p(u)] = \frac{7}{8}.$$

### Communication

Agreement with an allowed rule lets a receiver identify a damaged message.

### Cryptography

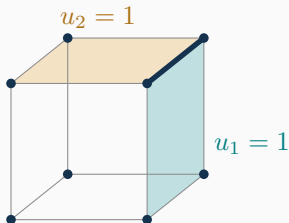
Agreement with an XOR of inputs makes a Boolean output predictable from that combination.

High algebraic degree alone does not prevent accurate affine prediction.

Carlet, Boolean Functions for Cryptography and Coding Theory. Component correlations are one design criterion, not a full cipher-security test.

# From recognizing a pattern to engineering an operation

The coordinate hyperplanes  $u_1 = 1$ ,  $u_2 = 1$ , and  $u_3 = 1$  meet on the same cube.



| function      | weight | geometry |
|---------------|--------|----------|
| $u_1$         | 4      | face     |
| $u_1 u_2$     | 2      | edge     |
| $u_1 u_2 u_3$ | 1      | vertex   |

XOR removes a shared one twice from the integer count:

$$\text{wt}(f + g) = \text{wt}(f) + \text{wt}(g) - 2 \text{wt}(fg).$$

Next: engineer these counts so that separate physical rotations give exactly the logical phase we want.

## A quantum phase is a function of a binary word

A computational basis state  $|v\rangle$  has one bit per physical qubit.

$$T = \begin{pmatrix} 1 & 0 \\ 0 & \omega \end{pmatrix}, \quad \omega = e^{i\pi/4}.$$

Applying  $T$  independently to every qubit gives

$$T^{\otimes n}|v\rangle = \omega^{\text{wt}(v)}|v\rangle.$$

The logical operation we want is

$$\text{CCZ}|a_1a_2a_3\rangle = (-1)^{a_1a_2a_3}|a_1a_2a_3\rangle.$$

Codeword weights determine relative phases. Only the logical label 111 should acquire a minus sign.

## Complementary codewords encode three quantum bits

Pair the affine tables with the same coefficient vector  $a$ :

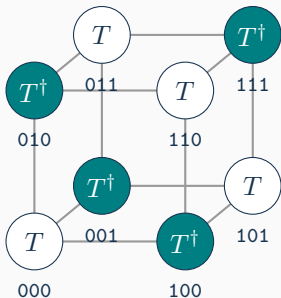
$$|a\rangle_L = \frac{|f_{a,0}\rangle + |f_{a,1}\rangle}{\sqrt{2}}.$$

| $a$ | $f_{a,0}$ | $f_{a,1}$ |
|-----|-----------|-----------|
| 000 | 00000000  | 11111111  |
| 100 | 00001111  | 11110000  |
| 111 | 01101001  | 10010110  |

Eight disjoint pairs give eight orthogonal encoded basis states.

The two strings in each pair must receive the same phase, or the gate leaves the encoded space.

## Parity chooses where to reverse the physical rotation



Use our original parity  $p(u)$  as the gate pattern:

$$p(u) = 0 : T, \quad p(u) = 1 : T^\dagger.$$

For a table  $v$ , count a one with sign  $(-1)^{p(u)}$ :

$$\text{sw}_p(v) = \sum_{u \in \mathbb{F}_2^3} (-1)^{p(u)} v(u).$$

The phase is  $\omega^{\text{sw}_p(v)}$ .

Four rotations and four inverse rotations act in one layer, with no gates between physical qubits.

## The gate check is another Hamming-distance calculation

At each position, as an integer,

$$(-1)^{p(u)}v(u) = (p+v)(u) - p(u).$$

Therefore

$$\text{sw}_p(v) = \text{wt}(p+v) - \text{wt}(p) = \text{wt}(p+v) - 4.$$

| logical label $a$  | $\text{sw}_p(f_{a,0})$ | $\text{sw}_p(f_{a,1})$ | phase |
|--------------------|------------------------|------------------------|-------|
| $a \neq (1, 1, 1)$ | 0                      | 0                      |       |
| $a = (1, 1, 1)$    | -4                     | 4                      |       |

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|--------------------|------------------------|------------------------|-------|
| $a \neq (1, 1, 1)$ | 0                      | 0                      | +1    |
| $a = (1, 1, 1)$    | -4                     | 4                      | -1    |

The distance calculation used for decoding now verifies a logical CCZ.

## A working gate still needs protection against phase errors

$$[[8, 3, 2]] \longrightarrow [[48, 3, 3]].$$

The entries mean physical qubits, logical qubits, and quantum distance.

Distance two detects an arbitrary single-qubit error; distance three corrects one.

A site's syndrome column lists the stabilizer-check responses to a phase error there.

Distinct nonzero columns ensure that no one or two columns sum to zero.

Choose the classical stabilizer code so that phase-error protection is built into the geometry.

Current paper, subspace-complement construction and the projectivity condition.

## The 48-site geometry gives divisible codewords

Keep  $\mathcal{P} = \{(u, w) \in \mathbb{F}_2^4 \times \mathbb{F}_2^2 : w \neq 0\}$ :  $64 - 16 = 48$  sites.

$w = (1, 0)$



16 points

$w = (0, 1)$



16 points

$w = (1, 1)$



16 points

Every nonzero linear rule has  
32 ones on the full space.

On the deleted  $w = 0$  subspace:  
8 ones if  $\alpha \neq 0$ , otherwise 0.

$$\ell(u, w) = \alpha \cdot u + \beta \cdot w, \quad \text{wt}(\ell|_{\mathcal{P}}) = \begin{cases} 32 - 8 = 24, & \alpha \neq 0, \\ 32 - 0 = 32, & \alpha = 0, \beta \neq 0. \end{cases}$$

The same half-space count gives stabilizer weights divisible by eight.

## Boolean label functions and signs complete the code

Choose physical sites



Assign logical labels



Choose rotation signs

$w = (1, 0)$

|             |             |             |     |
|-------------|-------------|-------------|-----|
| $T$         | $T$         | $T$         | $T$ |
| $T^\dagger$ | $T$         | $T^\dagger$ | $T$ |
| $T$         | $T^\dagger$ | $T^\dagger$ | $T$ |
| $T^\dagger$ | $T^\dagger$ | $T$         | $T$ |

6  $T^\dagger$

$w = (0, 1)$

|             |             |             |             |
|-------------|-------------|-------------|-------------|
| $T$         | $T^\dagger$ | $T$         | $T$         |
| $T$         | $T^\dagger$ | $T$         | $T$         |
| $T^\dagger$ | $T^\dagger$ | $T$         | $T^\dagger$ |
| $T$         | $T$         | $T^\dagger$ | $T$         |

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$w = (1, 1)$

|             |             |             |             |
|-------------|-------------|-------------|-------------|
| $T$         | $T^\dagger$ | $T^\dagger$ | $T^\dagger$ |
| $T^\dagger$ | $T^\dagger$ | $T$         | $T^\dagger$ |
| $T^\dagger$ | $T$         | $T$         | $T$         |
| $T^\dagger$ | $T^\dagger$ | $T$         | $T^\dagger$ |

10  $T^\dagger$

26  $T$  and 22  $T^\dagger$  implement logical CCZ on  $[[48, 3, 3]]$ ;  $d_X = 16$ ,  $d_Z = 3$ .

Current paper, fixed logical-label table and proposition giving the explicit sign polynomial.

## The paper reduces phase correctness to row overlaps

For the fixed signs, let  $\text{sw}_\gamma(g) = \sum_j (-1)^{\gamma_j} g_j$ . The exact conditions on distinct generator rows are

$$\text{sw}_\gamma(g) \equiv 0 \pmod{8},$$

Products denote coordinatewise intersections, just as on the cube.

Current paper, signed-overlap proposition; the block-weight identity extends the toy calculation.

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$$\begin{aligned}\text{sw}_\gamma(g) &\equiv 0 \pmod{8}, \\ \text{sw}_\gamma(gg') &\equiv 0 \pmod{4},\end{aligned}$$

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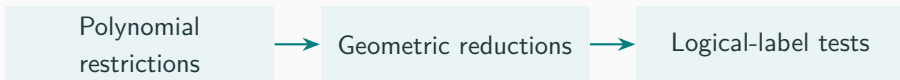
$$\begin{aligned}\text{sw}_\gamma(g) &\equiv 0 \pmod{8}, \\ \text{sw}_\gamma(gg') &\equiv 0 \pmod{4}, \\ \text{sw}_\gamma(gg'g'') &\equiv \begin{cases} 1 \pmod{2}, & \text{the three logical rows,} \\ 0 \pmod{2}, & \text{every other triple.} \end{cases}\end{aligned}$$

Products denote coordinatewise intersections, just as on the cube.

The cubic overlap supplies CCZ; the remaining conditions preserve each encoded superposition.

Current paper, signed-overlap proposition; the block-weight identity extends the toy calculation.

## Boolean structure also limits how short a code can be



Degree and weight restrict supports; geometric reductions and logical-label tests exclude candidates.

**Exclusion:** no  $n \leq 38$  CSS code with  $k = 3$ ,  $d_Z \geq 3$ , and the nine Campbell–Howard conditions.

The conditions allow a diagonal Clifford correction after physical  $T$  gates.

Lengths 39–46 remain open in this class. The 48-qubit construction does not establish global minimality.

Current paper, length-lower-bound theorem and hyperplane reductions.

## What the Boolean viewpoint made possible

**A construction:** 48 physical qubits protect three logical qubits; single-qubit phase rotations implement logical CCZ.

**An exact test:** signed codeword weights and overlaps certify the gate on every encoded state.

**A lower bound:** no  $n \leq 38$  CSS code with three logical qubits,  $d_Z \geq 3$ , and the nine Campbell–Howard conditions.

Boolean structure lets us recover a message, expose a predictor, and engineer a quantum gate.

## Backup: affine equivalence is a change of coordinates

Use *input-affine equivalence* to mean

$$g(u) = f(Au + t), \quad A \in \text{GL}(m, \mathbb{F}_2), \quad t \in \mathbb{F}_2^m.$$

The bijection  $u \mapsto Au + t$  permutes truth-table positions.

$$\text{wt}(g) = \text{wt}(f), \quad \deg g = \deg f \quad \text{for nonzero } f.$$

Under the same substitution for two functions, their distance is preserved. Derivatives transform as

$$D_v g(u) = D_{Av} f(Au + t).$$

A position permutation preserves the code when all codewords and physical signs are transported together. Translating syndrome entries is a different operation.

Convention: output-affine additions are not included in this definition.

## Backup: one error location represents all eight locations

Translate the received function by  $t = (1, 0, 0)$ :

$$r_t(u) = r(u + t) = p(u) + 1 + \delta_t(u).$$

The error moves from 000 to 100 and the original parity becomes its complement.

$$\text{RM}(1, 3) \xrightarrow{u \mapsto Au+t} \text{RM}(1, 3).$$

Every invertible affine substitution preserves the whole Reed–Muller family and its Hamming distances.

Relabeling reduces duplicate cases in a search and transports a decoding argument to another coordinate system.

## Backup: input equivalence and output addition differ

A broader convention, often called extended-affine equivalence, allows

$$g(u) = f(Au + t) + \ell(u), \quad \deg \ell \leq 1.$$

Adding  $\ell$  preserves distance to the affine code, but need not preserve weight:

$$f = 0, \quad \ell = u_1 \quad \implies \quad \text{wt}(f) = 0, \quad \text{wt}(f + \ell) = 4 \quad (m = 3).$$

In the paper's punctured support formula  $n = \text{wt}(c) - c(0)$ , a translation can also change which value is deleted at the origin.

A classification must specify its equivalence relation and track the distinguished puncture.

Current paper, admissible-block-length theorem and normal-form reductions.

## Backup: a translated hyperplane can change syndrome rank

Translate the four-point face  $z = 1$  by 001:

$$\{001, 011, 101, 111\} \longrightarrow \{000, 010, 100, 110\}.$$

$$\begin{pmatrix} 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 \end{pmatrix} \longrightarrow \begin{pmatrix} 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

Both sets have affine dimension two; their column ranks are three and two.

The translated origin has zero syndrome. A nonzero logical label there would still give a nonzero full generator column.

The example illustrates syndrome geometry; translation alone does not guarantee preservation of quantum distance.

## Backup: a polynomial factor exposes a geometric reduction

In the paper's rank-eight candidate decomposition,  $f = 0$  gives

$$c = y_8(g + y_7h) \implies \text{supp}(c) \subseteq \{y_8 = 1\}.$$

After choosing the origin, the containing hyperplane either:

- passes through zero, so actual rank is at most seven; or
- avoids zero and has a missing support point, supplying a projection direction.

In the origin-avoiding rank-eight case,  $n \leq 38 < 128$  leaves a missing point  $v$ : no supported point or supported pair sum equals  $v$ .

For that case, the paper's CH,  $k = 3$ ,  $d_Z \geq 3$ , and distinct-nonzero-syndrome hypotheses permit projection to smaller rank at the same length.

Current paper, hyperplane reduction and rank-eight exclusion; lower ranks have already been excluded.

## Backup: derivatives recover ANF coefficients

For  $S \subseteq \{1, \dots, m\}$ , write  $u_S = \prod_{i \in S} u_i$ . Then

$$f(u) = \sum_{S \subseteq \{1, \dots, m\}} c_S u_S, \quad c_S = \left( \prod_{i \in S} D_{e_i} \right) f(0).$$

All arithmetic is in  $\mathbb{F}_2$ . Every derivative of order greater than  $\deg f$  vanishes.

$$D_{e_1} D_{e_2} D_{e_3} r = 1.$$

On three variables, the third derivative is the sum of all eight values: a parity check for  $\text{RM}(2, 3)$ .

Higher derivatives turn polynomial degree constraints into local consistency tests.

## Backup: why low degree forces nonzero weight

Write  $f(u', u_m) = g(u') + u_m h(u')$ , with  $\deg g \leq d$  and  $\deg h \leq d - 1$ . The two halves of the truth table are  $g$  and  $g + h$ .

$$\text{wt}(f) = \text{wt}(g) + \text{wt}(g + h) \geq \text{wt}(h).$$

If  $h \neq 0$ , induction gives  $\text{wt}(h) \geq 2^{(m-1)-(d-1)} = 2^{m-d}$ . If  $h = 0$ , the two halves agree and induction gives the same bound.

$$\text{RM}(d, m) : \left[ 2^m, \sum_{j=0}^d \binom{m}{j}, 2^{m-d} \right], \quad 0 \leq d \leq m.$$

The monomial  $u_1 \cdots u_d$  attains the bound; the case  $d = m$  has the trivial lower bound one.

## Backup: degree trades information rate for protection

Evaluate all degree-at-most- $d$  polynomials on the same eight points.

| degree bound $d$ | classical code | protection               |
|------------------|----------------|--------------------------|
| 0                | $[8, 1, 8]$    | correct any three errors |
| 1                | $[8, 4, 4]$    | correct any one error    |
| 2                | $[8, 7, 2]$    | detect any one error     |
| 3                | $[8, 8, 1]$    | every word is allowed    |

More monomials carry more information, but bring the allowed messages closer together.

## Backup: Fourier analysis is another decoder

Represent the table by real signs and correlate with a parity:

$$W_r(a) = \sum_{u \in \mathbb{F}_2^3} (-1)^{r(u) + a \cdot u}.$$

The score is agreements minus disagreements, so

$$d(r, f_{a,b}) = \frac{8 - (-1)^b W_r(a)}{2}.$$

For our received word,  $W_r(111) = 6$  and every other score is  $-2$ .

The largest absolute correlation finds the nearest affine word; the sign determines its constant term.

Walsh/Fourier viewpoint: O'Donnell, Analysis of Boolean Functions.

## Backup: inclusion–exclusion exposes the cubic term

For binary functions, the following identity holds over the integers:

$$\begin{aligned}\text{wt}(f + g + h) = & \text{wt}(f) + \text{wt}(g) + \text{wt}(h) \\ & - 2[\text{wt}(fg) + \text{wt}(fh) + \text{wt}(gh)] + 4\text{wt}(fgh).\end{aligned}$$

The same identity holds for signed weight, because the identity holds at every coordinate before summing.

For sums of more rows, an intersection of  $j$  rows carries coefficient  $(-2)^{j-1}$ .

Modulo eight, intersections of four or more rows disappear. Row, pair, and triple tests determine the phase.

## Backup: why the toy quantum distance is two

Let  $C_1 = \text{RM}(1, 3)$  and  $C_2 = \langle \mathbf{1} \rangle$ . The encoding pairs words in each coset of  $C_2$  inside  $C_1$ .

$$d_X = \min_{v \in C_1 \setminus C_2} \text{wt}(v) = 4.$$

A vector supported on sites 000 and 100 has even weight, but overlaps  $f_{100,0}$  in one position. Therefore

$$d_Z = \min_{v \in C_2^\perp \setminus C_1^\perp} \text{wt}(v) = 2, \quad d = \min(d_X, d_Z) = 2.$$

Classical distance protects against one kind of error; quantum distance must also account for the dual constraints.

## Backup: the full signed-weight and block tests

For generator rows  $G = [K; S]$ , an encoded basis state superposes

$$c(a, y) = K^T a + S^T y.$$

A native sign pattern must satisfy

$$\text{sw}_\gamma(c(a, y)) \equiv 4a_1a_2a_3 \pmod{8} \quad \text{for all } a, y.$$

The paper's block criterion is the exact integer identity

$$\text{sw}_\gamma(c) = \sum_b [\text{wt}(\gamma_b + c_b) - \text{wt}(\gamma_b)].$$

Once the signed-overlap conditions involving stabilizer rows hold, only the eight logical inputs at  $y = 0$  need to be checked.

Current paper, signed-overlap and block-criterion propositions.

## Backup: the 48-qubit sign polynomial

On  $(u, w) \in \mathbb{F}_2^4 \times (\mathbb{F}_2^2 \setminus \{0\})$ ,

$$\gamma(u, w) = u_3 + w_1 Q_1(u) + w_2 Q_2(u),$$

$$Q_1(u) = u_1 + u_2 + u_3 + u_1 u_3 + u_2 u_3 + u_2 u_4,$$

$$Q_2(u) = u_3 + u_4 + u_1 u_2 + u_1 u_4 + u_3 u_4.$$

All sums in these polynomials are in  $\mathbb{F}_2$ . Each restriction to a fixed  $w$  is quadratic; the block weights are 6, 6, 10.

The polynomial specifies the physical rotations; the fixed logical rows and signed-weight test establish the gate action.

Current paper, explicit sign-pattern proposition. Diagram rows:  $u_1 u_2$ ; columns:  $u_3 u_4$ , both ordered 00, 01, 11, 10.

## Backup: how short can the code be?

The paper considers CSS codes with  $k = 3$ ,  $d_Z \geq 3$ , and the nine Campbell–Howard conditions.

| length              | result in the paper                                |
|---------------------|----------------------------------------------------|
| $n \leq 38$         | no code in this class, at any stabilizer rank      |
| $39 \leq n \leq 46$ | existence remains open                             |
| $n = 47$            | the cited code has no native $T/T^\dagger$ pattern |
| $n = 48$            | the constructed code has a native pattern          |

For projective support of stabilizer rank  $s$ ,

$$\chi_S \in \text{RM}(s-4, s) + \langle \delta_0 \rangle, \quad n = \text{wt}(c) - c(0), \quad c \in \text{RM}(s-4, s),$$

and the nonzero support must span  $\mathbb{F}_2^s$ .

The 47-qubit statement concerns that particular code. The 48-qubit construction is not a proof of global minimality.

## Sources for this talk

**Research results:** the current paper, *A 48-qubit native CCZ code from projective geometry and a length lower bound*.

- Subspace-complement construction and fixed logical labels.
- Signed-overlap criterion, block-weight identity, and explicit sign pattern.
- Reed–Muller support restrictions and the conditional length lower bound.

**Fundamentals:** MacWilliams–Sloane and Huffman–Pless for classical codes; Carlet for ANF, derivatives, affine prediction, and equivalence.

**Spectral viewpoint:** O'Donnell, *Analysis of Boolean Functions*.